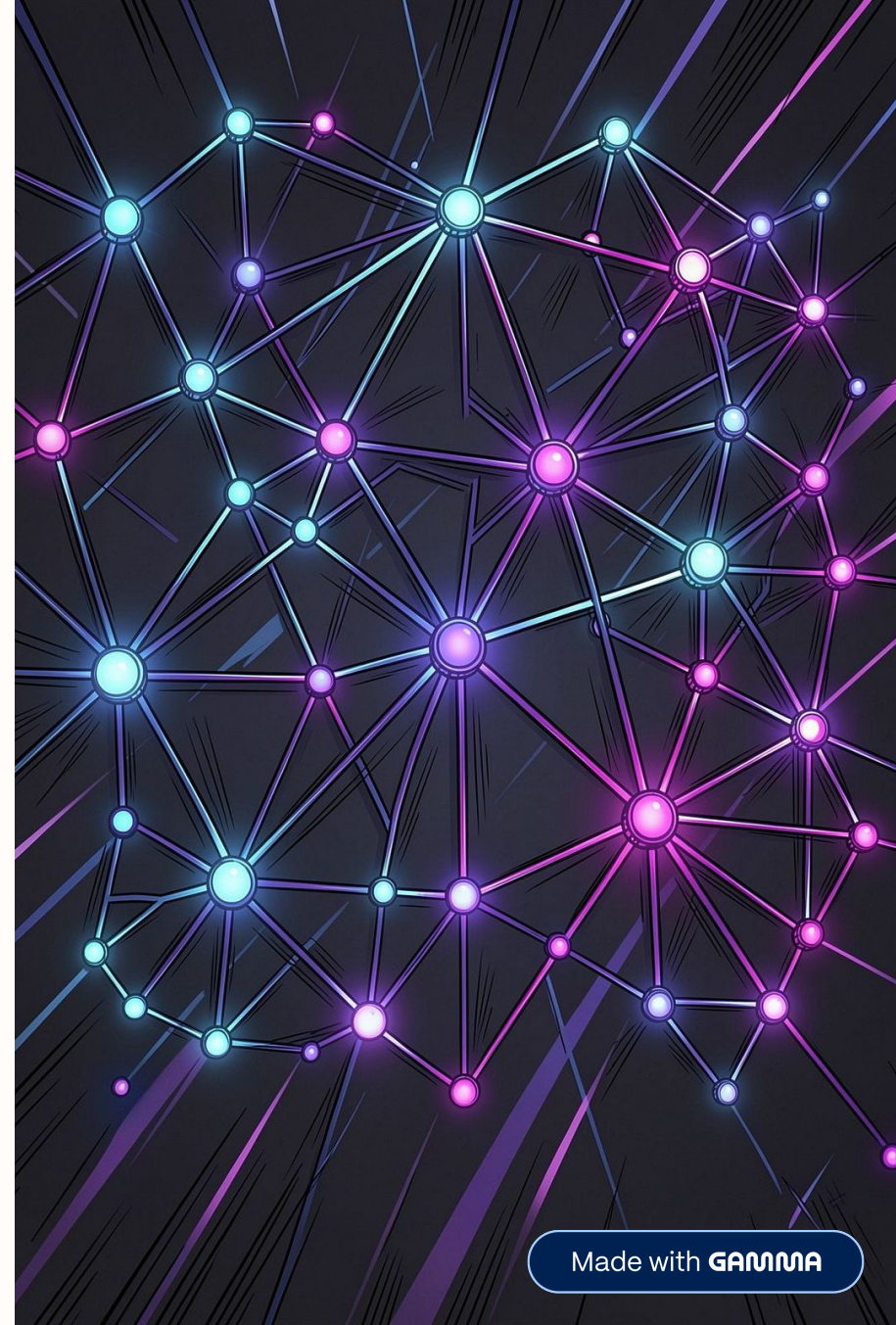


ThetaSwarm: Autonomous Options Alpha Desk

The AI hedge fund that separates intelligence from execution.

4 Specialized Agents. Mathematical Guardrails. Zero Human Emotion.



Made with GAMMA

Why Most AI Trading Agents Fail

Giving an LLM the keys to your brokerage account is a recipe for disaster.

The "Black Box" Problem

Most AI bots invent trades based on hallucinations — without showing their math.

Zero Risk Discipline

An LLM reads news well but cannot manage portfolio exposure or price options correctly.

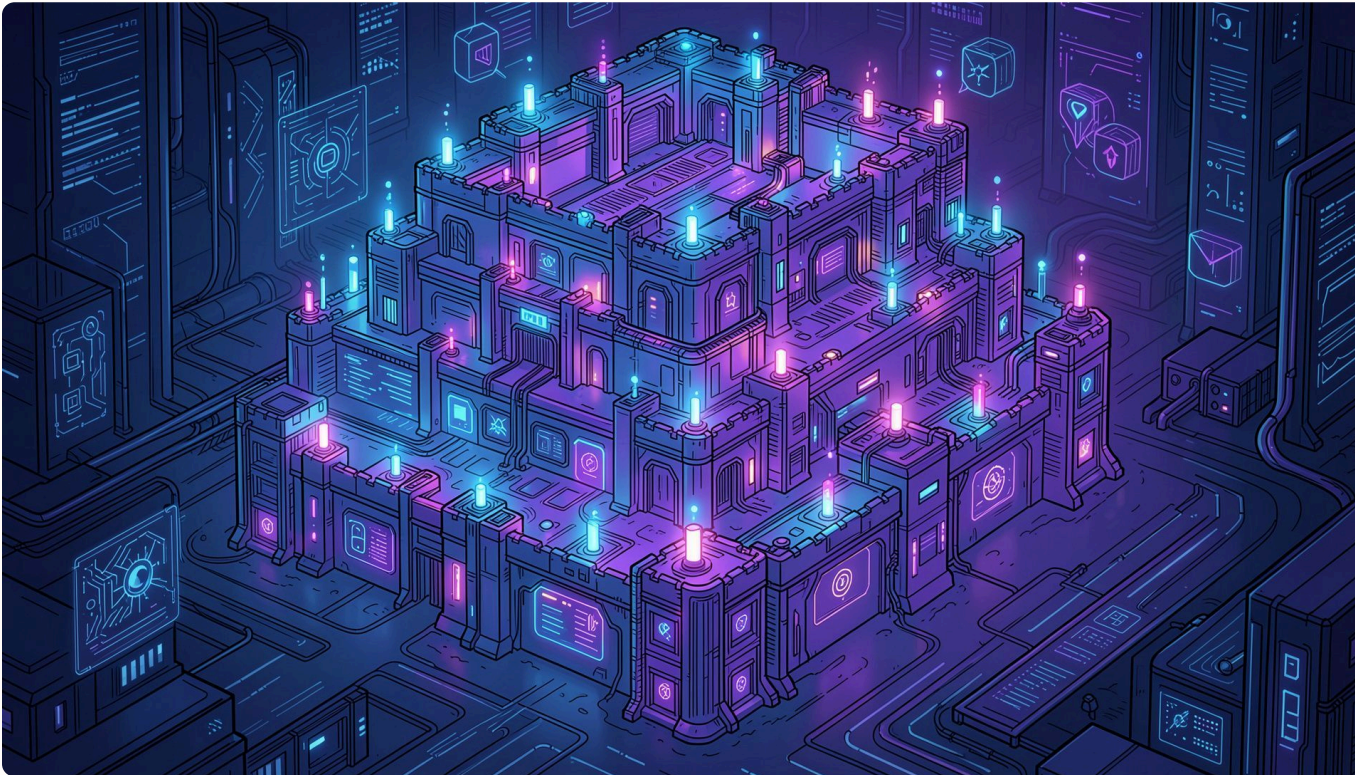
The Retail Disadvantage

Emotional decision-making and delayed execution during macro shocks destroy retail traders. The solution: let AI *propose*, and let deterministic code *decide*.



The "Fail-Closed" Architecture

Intelligence is Autonomous. Execution is Governed. ThetaSwarm is built around one strict principle: **Separation of Powers**.



The AI Swarm

Qwen 2.5 72B reads market news, calculates Breeden-Litzenberger probabilities, and proposes options strategies.

The Adversarial Critic

Red-teams every trade, actively searching for reasons to veto it before execution.

Hard-Coded Risk Engine

Six deterministic mathematical gates. Fail one, and the trade is killed instantly.

From Market Signal to Live Alpaca Execution

01

Market Data Poller

Ingests live news and Alpaca market snapshots every 3 minutes.

02

Narrative Agent

Reads the news and predicts Implied Volatility — Expansion vs. Crush.

03

Quant Agent

Fetches live option chains and structures a defined-risk multi-leg trade (e.g., Long Straddle).

04

Critic & Risk Manager

The Critic attacks the thesis. The Python Risk Manager checks buying power and max loss limits.

05

Alpaca MCP Execution

If approved, atomic multi-leg orders fire directly to Alpaca Paper Trading via the Model Context Protocol.

A Trading Agent You Can Actually Audit

Good trading is about the trades you *don't* take.



Predictive Alpha Backtest

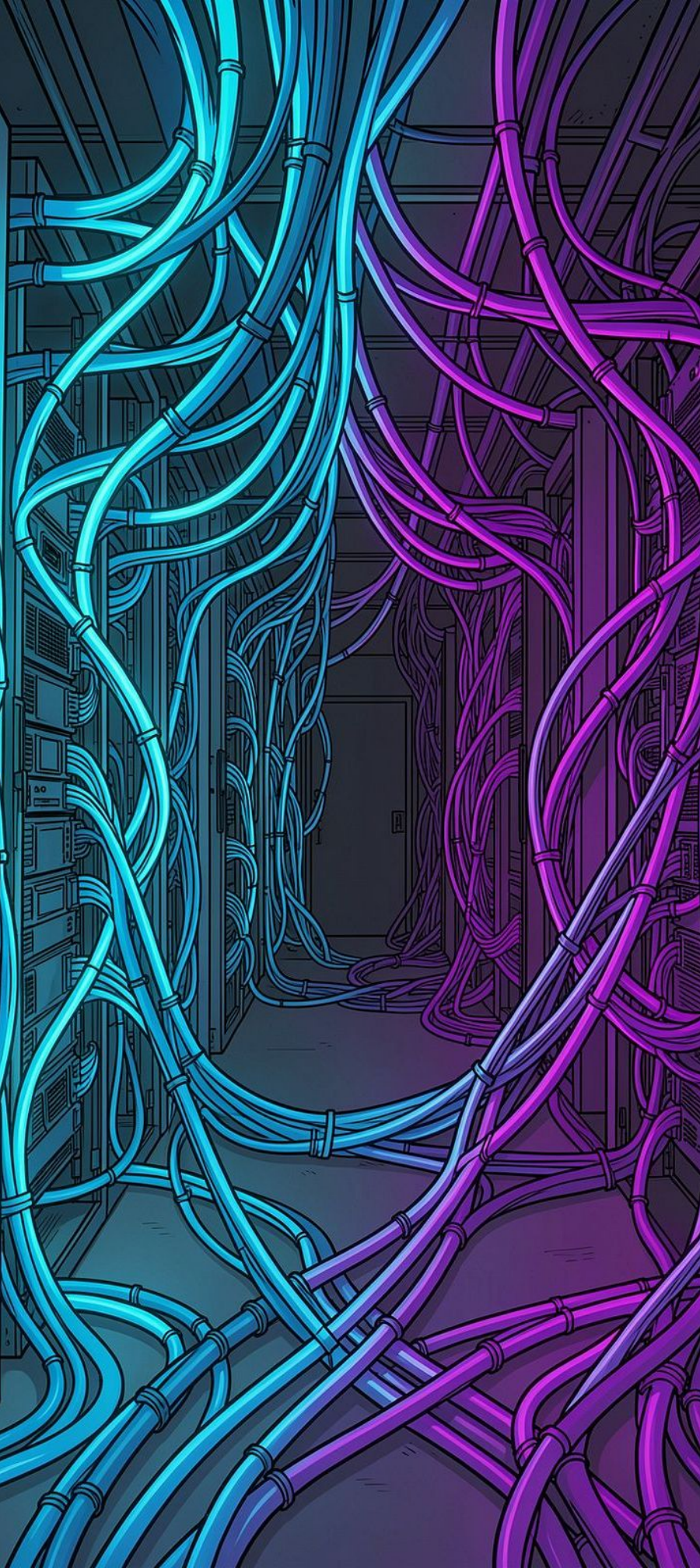
Against the 5 largest macro shocks of 2024, our Live Swarm Episodic Backtester structured options trades that captured volatility — profiting when the market panicked.

The Shadow Book

An append-only cryptographic ledger of every trade the AI *refused* to take, visible on our live Streamlit dashboard.

Real Example

Day 1: Quant Agent proposed a Bull Put Spread. The Critic rejected it — IV thesis misaligned, saving capital from a mathematically poor setup.



TECH STACK

Built on Modern, Institutional-Grade Infrastructure



Backend

Python, SQLite for audit logging, and CrewAI for agent orchestration.



AI Brain

Qwen-2.5-72B-Instruct via Featherless AI — sub-second, model-agnostic reasoning.



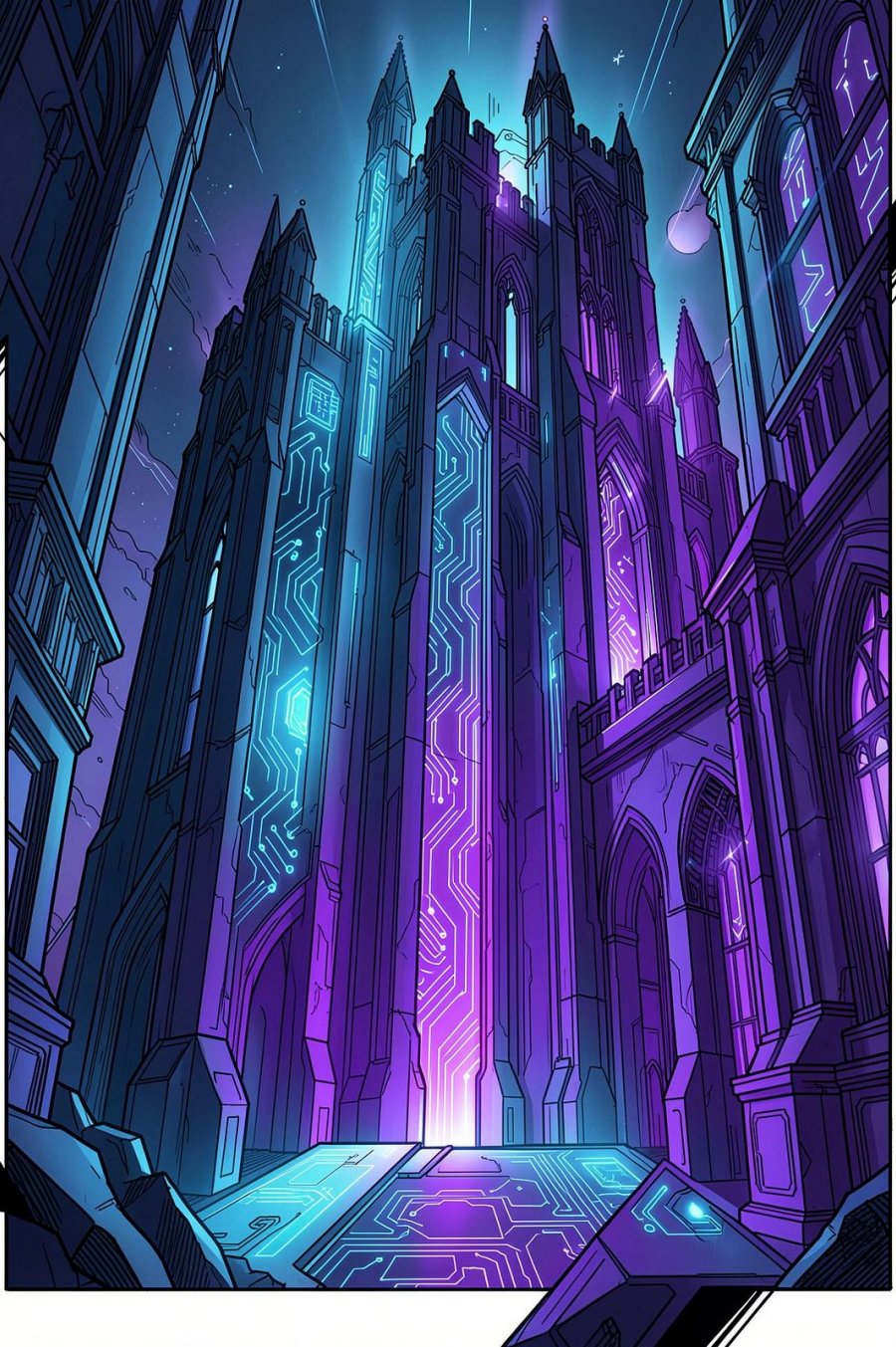
Broker & Data

Alpaca Markets API and Alpaca MCP Server for secure, locked-down order execution.



Frontend

Bloomberg-style, glassmorphic Streamlit terminal showing live Agent Thoughts, Portfolio Greeks, and active executions.



TRACK: OPTIONS ALPHA AGENTS

Ready for Judge Verification

Transparent. Reproducible. Risk-Aware.

End-to-End Verified

Full execution verified on
Alpaca Paper Servers.

Real Multi-Leg Contracts

Options contracts generated
autonomously — no simulation
shortcuts.

Zero Fabrication

No fake fills. No hidden curve-fitting. Every decision is auditable.

We didn't just build a trading bot. We built the fortress around it.